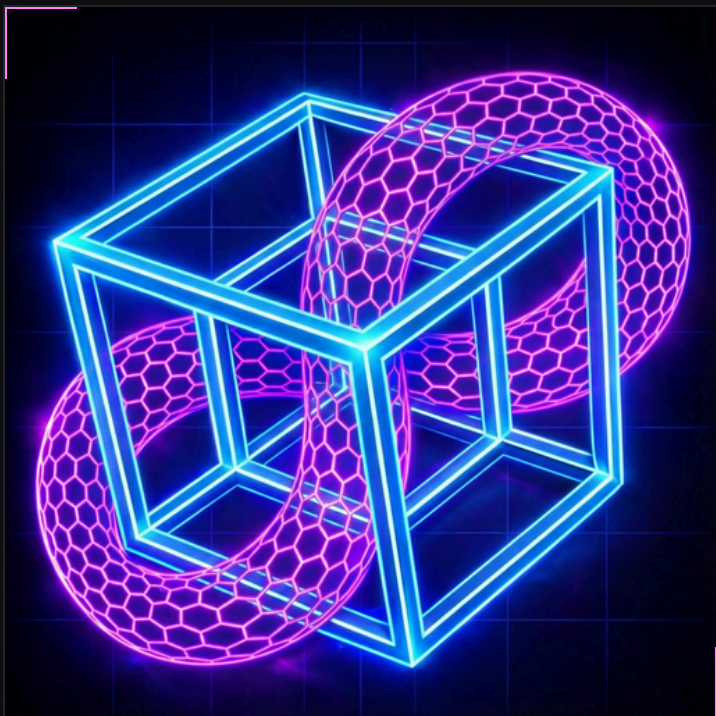




PolyStore

Decentralized Storage and Delivery

Decentralized storage has focused on keeping data.
PolyStore also delivers data.

mike@polynomialstore.com

Stored Does Not Mean Delivered.

Decentralized storage networks can prove that data is stored, but reliable delivery remains weakly enforced.

Decentralized storage networks do not guarantee reliable retrieval.

PolyStore Proves Delivery.

PolyStore uses efficient KZG Polynomial Proofs to cryptographically prove storage and retrievals on-chain.

PolyStore's protocol holds storage providers accountable for both storing and delivering data.

PolyStore Software Stack

polynomialstore.com/#/dashboard

Browser Client

/DASHBOARD

Dashboard

SHOW DIAGNOSTICS

/REGISTRY/DEALS

Select container to manage files.

+ NEW DEAL

DEAL FILTER

Search by id or manifest root

DEAL

STATUS

SIZE

LAST ACTIVITY

#2

ACTIVE

53.9 MiB

/DEAL/EXPLORER

Deal #2

RS(2,1)

53.92 MB

ACTIVE

0x8fbb32fa4d5edfc48c8941875713074b1d6db2ef0a91149b905bf0dc9954ebfd9b88ed4dd49b46ae3cd56875d747184a

/DEAL/UPLOAD

1. SELECT FILE

PROCESSING IN BROWSER

SELECT FILE

PolyStore Desktop Client

LOCAL GATEWAY

OPEN DASHBOARD

READY

LISTENING ON 127.0.0.1:8080

3/3 CHECKS READY

SYSTEM HEALTHY

SYNCH 11:20:33 AM

OVERVIEW

STORAGE

DIAGNOSTICS

GATEWAY PROCESS

READY

GUI-managed local process

CHAIN CONNECTIVITY

READY

LCD reachable

STORAGE PROVIDERS

READY

Provider paths reachable

RECENT ACTIVITY

DEAL STORAGE LAYOUT (SLAB / MDUS)

MDU education lives on the Technology page. This panel focuses on visibility and developer-grade ranges.

MANIFEST ROOT

Deal ID / Owner

123

MANIFEST ROOT

CONSTANTS

mdu_size=8,388,608

blob_size=131,072

SLAB BREAKDOWN

meta=1

witness=2

user=7

total=7

WITNESS OVERRIDE (OPTIONAL)

Terminal · CLI Client

\$ nil-cli shard dataset.tar

Sharding file: "dataset.tar"

Processing MDU 1/4...

Computing Manifest...

PolyFS Root: 0x92f98d4f...

Daemon · Protocol Node

\$ polystorechaind start

CometBFT · block production active

RPC 127.0.0.1:26657

LCD 127.0.0.1:1317

EVM RPC 127.0.0.1:8545

Daemon · Storage Provider

persona provider-daemon

endpoint 127.0.0.1:8091

provider #1 registered

shard service ready

retrieval path healthy

Shared Core

Rust

PolyFS

Erasure coding

KZG commitments

WASM + native

One shared core spans clients, protocol nodes, and storage providers.

PolyStore Scales

BASE CHAIN

~22M
retrieval sessions/day

CHANNELS

~22B
retrieval sessions/day

1,000× modeled capacity

SUBCHAINS

~10T
deliveries/day

450× modeled parallel capacity
Two recursive layers

Modeled capacity approaches Cloudflare and Amazon S3 request volumes.

Founder / Protocol Lead



Mike Seiler

Cornell Applied Physics
BS · M.Eng.

**Protocol
Contributions**

Filecoin EVM compatibility FIPs, Ethereum EIP-918.

**Startup + Product
Execution**

Founding lead developer at Moat, acquired by Oracle in a reported \$850M transaction. CTO and co-founder of Torch, a social media research and intelligence platform for government, political campaigns, and nonprofits.

Start With Decentralized Storage Users

DECENTRALIZED STORAGE MARKET

Existing Market for Decentralized Storage

Existing Filecoin, Arweave, IPFS, and Storj users, traditional/centralized organizations and decentralized protocols already encountering failed retrievals.

CLOUD HYPERSCALER MARKET

Distributed Object Storage Market

Cloud data storage market with decentralized global scale.

AI NATIVE MARKET

AI Agent Storage Market

PolyStore data storage to support the AI Economy.

Start with users already experiencing failed retrievals, then expand to broader production workloads.

Build the Early Community

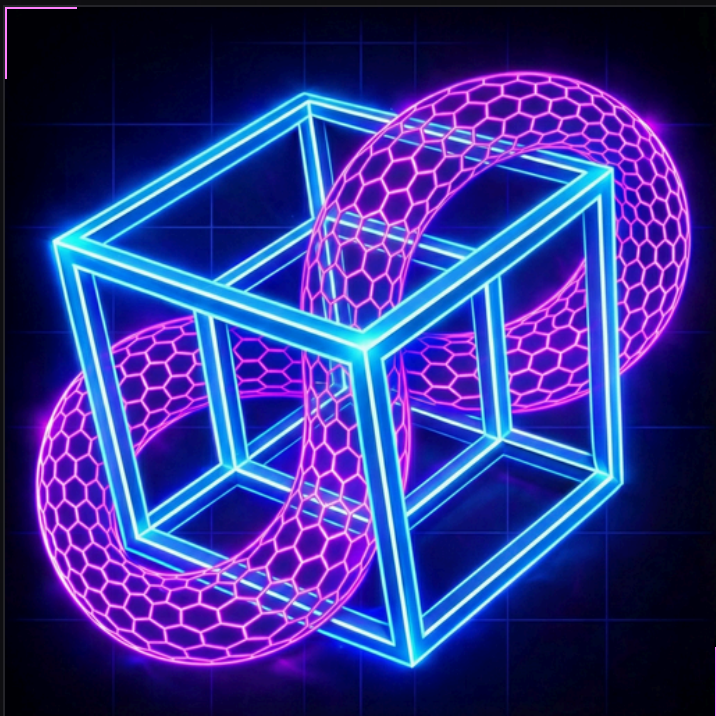
Workloads Real datasets with retrieval needs

Operators Providers and gateways running on the testnet

Team Prospective co-founders

Funding Aligned early investors

Early market development with the right people involved.



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How a PolyStore Deal Delivers Data

1 FUND THE DEAL

Reserve funding for storage and retrieval.

2 COMMIT THE DATA

PolyFS anchors the exact file state.

3 ASSIGN PROVIDERS

Place erasure-coded shards across provider slots.

4 SERVE + VERIFY

Serve requested bytes and verify them against the committed PolyFS root.

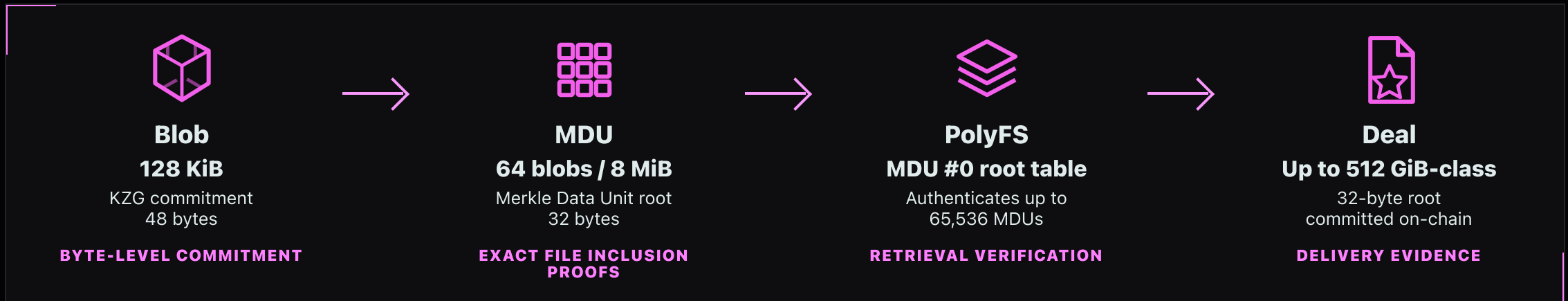
5 PAY OR REPAIR

Verified delivery earns payment. Repeated failure changes placement.

Fund storage and retrieval, commit the file state, verify delivery, then pay or repair.

PolyFS Primer

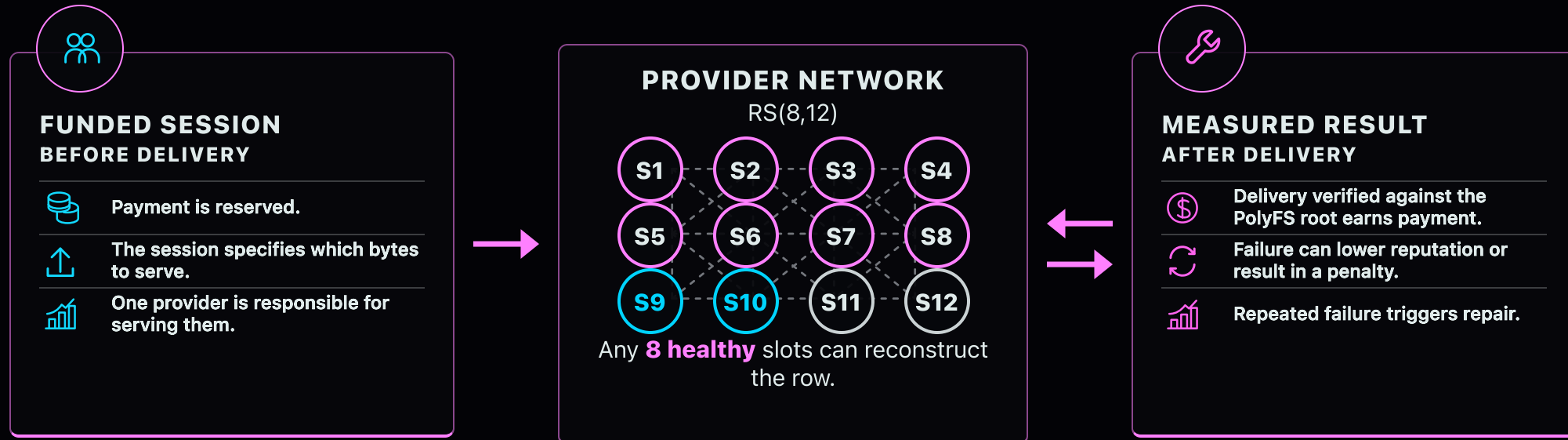
PolyFS anchors each Deal generation with a **32-byte root** committed on-chain. Off-chain proofs bind retrieved bytes to that state.



One 32-byte root anchors the file state used to verify retrievals.

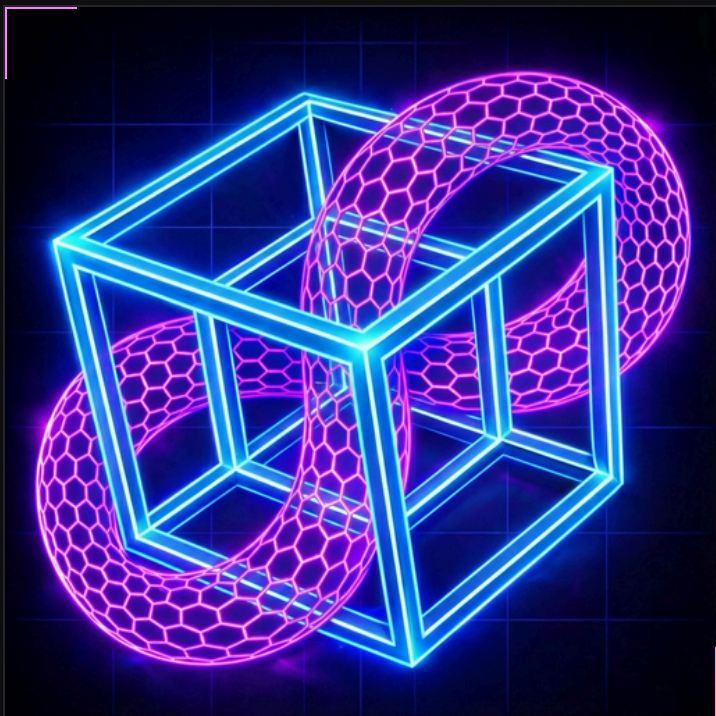
Verified Delivery Earns Payment

Repeated failure can lower reputation, result in penalties, and trigger repair.



○ Healthy (8) ○ Repair (2) ○ Unavailable (2)

Payment, reputation, penalties, and repair follow measured results — not self-reported uptime or unverifiable claims of service.



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